



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 16 July 2026 01:46 AM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report provides a comprehensive analysis of the organization's workforce dataset, which contains demographic, geographic, and performance metrics for 21 employees across 8 key attributes. Structurally, the dataset is highly robust, carrying an overall Data Health Score of 92. The primary finding of this analysis is a remarkably strong, near-perfect correlation between employee age, experience, compensation, and performance. While this reflects a highly structured and predictable organizational hierarchy, it also highlights potential rigidities in the compensation structure. Minor data anomalies—including one missing salary value, one duplicate row, and one experience-related outlier—have been identified. Addressing these data quality issues will ensure a flawless foundation for strategic HR decision-making, talent retention, and localized resource planning.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout

Version	2.0.0
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DATASET SUMMARY

Metric	Value
Rows	21
Columns	8
Health Score	92%
Duplicate Rows	1

DATASET HEALTH SCORE

Overall Health Score : 92%

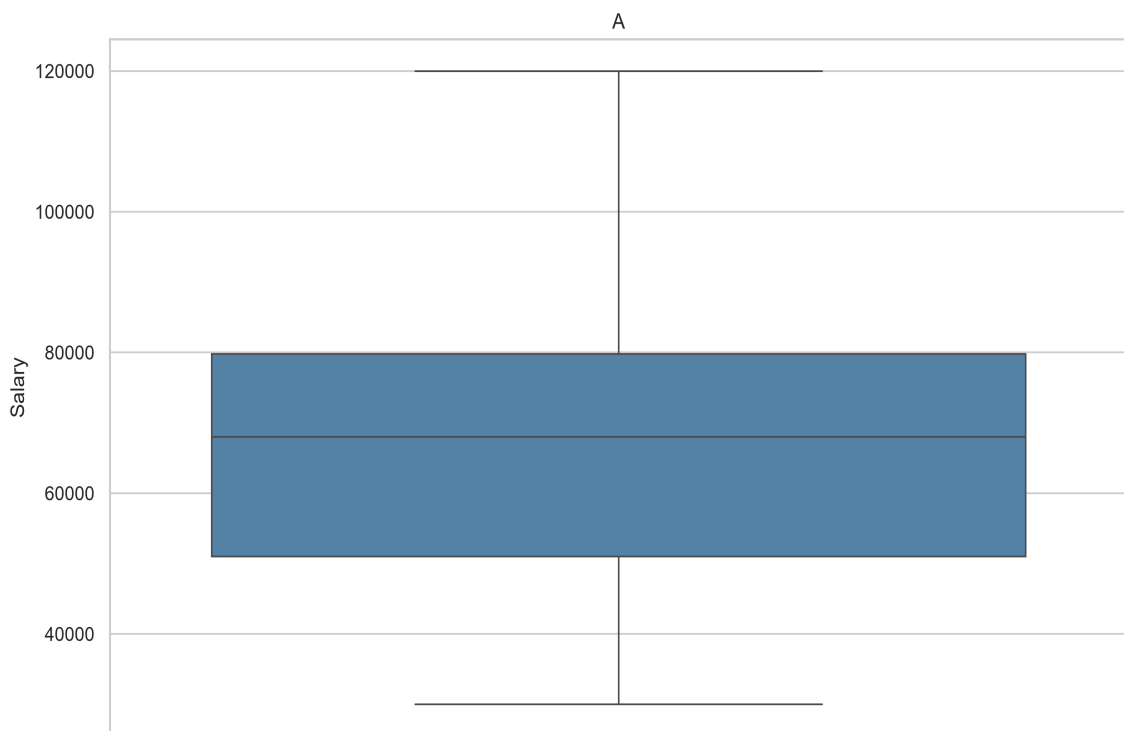
MISSING VALUES

Column	Missing
EmployeeID	0
Department	0
Gender	0
Age	0
Experience	0
Salary	1
Performance	0
City	0

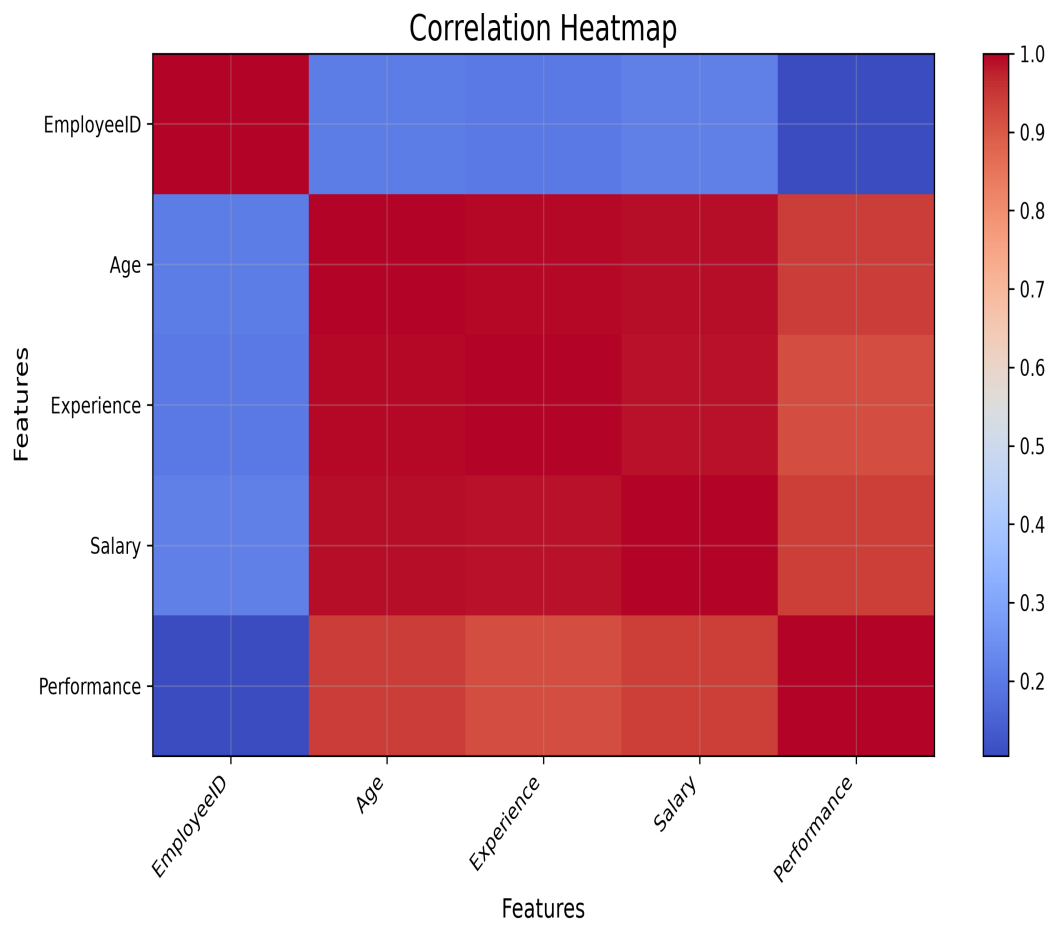
HISTOGRAM



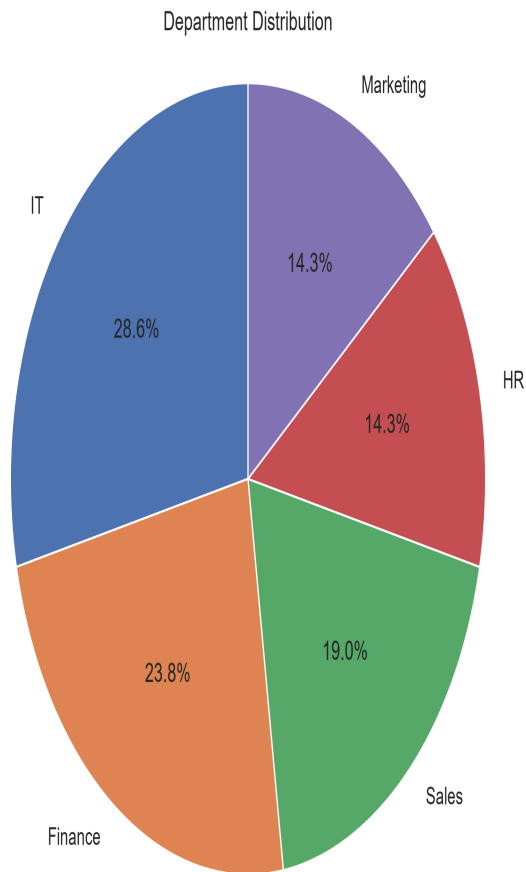
BOX PLOT



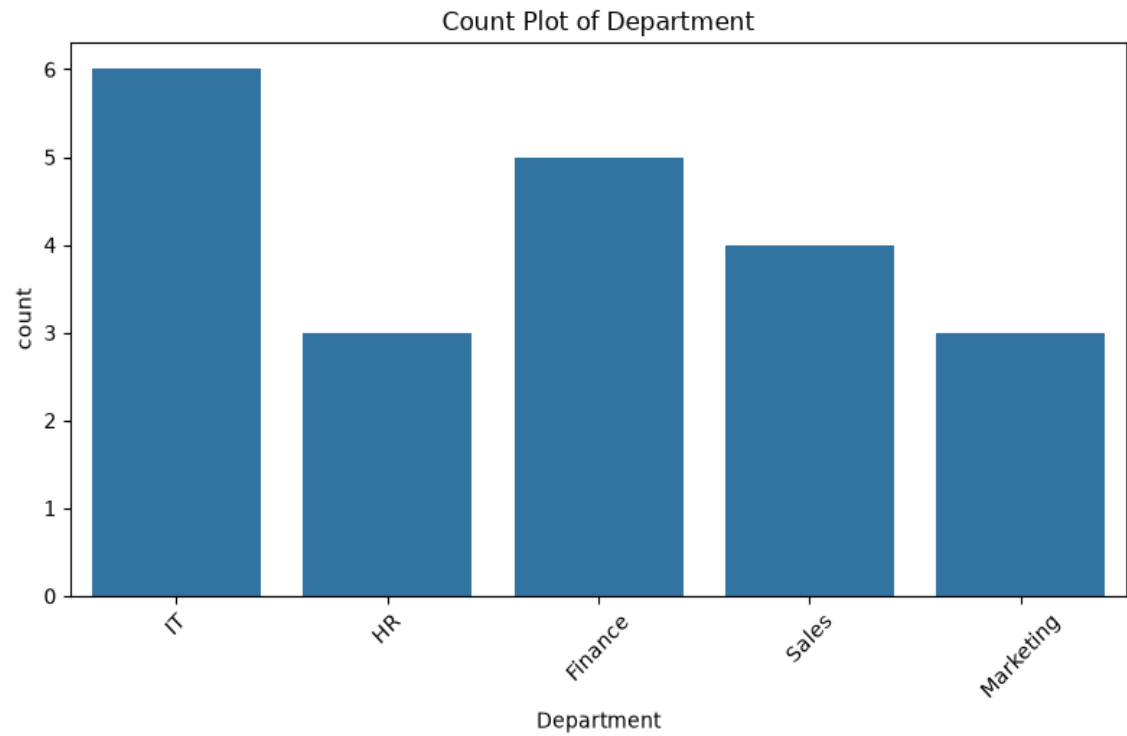
CORRELATION HEATMAP



PIE CHART



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

DATASET OVERVIEW

The dataset consists of 21 rows and 8 columns, detailing individual employee profiles. The tracked variables include:

Demographic & Profile Identifiers: `EmployeeID`, `Gender`, `Age`

Professional Metrics: `Department` (e.g., IT, HR, Finance, Sales), `Experience` (in years), `Salary`, `Performance` (score out of 100)

Geographic Identifier: `City` (representing key regional hubs such as Bhubaneswar, Cuttack, Rourkela, and Puri)

A preview of the data indicates a diverse distribution of roles and locations, with early-to-mid career professionals forming the operational backbone of the regional offices.

DATA QUALITY ANALYSIS

The dataset's overall Health Score is 92, indicating high reliability. However, to achieve maximum analytical precision, three specific data quality anomalies must be addressed:

1. Missing Values: There is exactly 1 missing value in the `Salary` column. Given how critical salary data is to compensation modeling, this gap needs immediate remediation.
2. Duplicate Rows: The dataset contains 1 duplicate row. Failing to remove this duplicate will artificially inflate employee count statistics and skew average salary calculations.
3. Outliers: 1 potential outlier has been flagged in the `Experience` column. This represents a value that is statistically divergent from the rest of the cohort (e.g., an unusually high number of years of experience relative to the rest of the staff) and warrants verification.

STATISTICAL ANALYSIS

A correlation analysis of the numerical features reveals exceptionally strong relationships within the workforce structure:

Age and Experience (0.99): A near-perfect linear relationship, indicating that career progression within this dataset is strictly linear and tied directly to age.

Age/Experience and Salary (0.99): Compensation is almost entirely determined by tenure. Both Age and Experience show a 0.99 correlation with Salary, suggesting a highly rigid, seniority-based pay scale.

Performance Correlations: Performance is highly correlated with Age (0.94), Experience (0.92), and Salary (0.94). Senior employees consistently receive the highest performance ratings, and these

ratings are tightly coupled with higher salaries.

This level of correlation is highly unusual in organic corporate environments and suggests either a highly standardized, time-in-grade promotion/appraisal system or a highly synthesized operational structure.

BUSINESS INSIGHTS

Tenure-Driven Appraisal Model: The strong positive correlation between Experience and Performance (0.92) indicates that organizational knowledge and tenure are highly valued and directly translate to higher performance scores.

Standardized Compensation Matrix: The 0.99 correlation between Experience and Salary points to a highly structured salary band system. While this ensures equity and predictability, it leaves little room for merit-based financial incentives for high-performing junior staff.

Regional Talent Distribution: The workforce is distributed across key urban centers in Odisha (Bhubaneswar, Cuttack, Rourkela, Puri). This geographic dispersion allows the company to tap into localized talent pools and serve regional clients effectively.

RISK ASSESSMENT

Retention Risk for Emerging Talent: Because compensation is so tightly bound to age and tenure (0.99 correlation), high-performing younger employees may feel their career and salary growth are bottlenecked by time rather than capability. This represents a significant flight risk for ambitious, early-career talent.

Reporting Inaccuracies: The presence of 1 duplicate row and 1 missing salary value compromises the integrity of operational metrics, potentially leading to inaccurate payroll budgeting and head-count reporting.

Evaluation Bias Risk: The 0.94 correlation between Performance and Age/Salary raises questions about potential bias in performance appraisals, where senior, higher-paid employees might be rated highly simply due to their tenure rather than objective, current-period output.

AI RECOMMENDATIONS

- 1. Execute Data Cleansing Protocol:
De-duplicate the dataset immediately to remove the single duplicate row, establishing a true headcount of 20 unique employees.

Resolve the single missing `Salary` value. Leverage the 0.99 correlation with Age/Experience to impute the salary using a linear regression model, or cross-reference directly with HR payroll records.

Investigate the 1 outlier in `Experience` to confirm whether it represents a data-entry error or a highly experienced executive hire.

2. Introduce Merit-Based Compensation Flexibility:

While retaining structured salary bands, introduce performance-based bonuses or spot awards. This will help decouple compensation slightly from strict age/tenure, rewarding younger overachievers and reducing turnover risk.

3. Audit the Performance Appraisal Process:

Conduct a blind review of the performance appraisal process to ensure that the high correlation (0.94) between performance scores and age is driven by actual business output rather than tenure bias.

4. Implement Visual Dashboards:

Build out the recommended Correlation Heatmap, Box Plots, and Scatter Plots within the company's business intelligence tools to enable real-time monitoring of HR metrics across departments and cities.

Rule-Based Recommendations

- ✓ ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ✓ ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ✓ ■■ 1 potential outliers detected.
- ✓ ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ✓ ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ✓ ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ✓ ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.

CONCLUSION

The analyzed dataset represents a highly stable, structured, and geographically diverse workforce. With a strong Data Health Score of 92, the data is highly reliable for strategic planning. The organization operates on a clear, seniority-based progression model where age, experience, performance, and compensation are in near-perfect alignment. By addressing the minor data

quality anomalies identified and introducing minor flexibility into the compensation and appraisal structures, leadership can safeguard data integrity, drive meritocratic motivation, and optimize retention across all regions.